

CS 4510 Automata and Complexity

Exam Practice 1

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I would consider these problems very difficult, and not representative of the difficulty of the exam. Your exam may not cover some of the topics listed here; it is a super set.

1. Prove that an infinite union (as opposed to a finite one) of regular languages is not necessarily regular.
2. We say a homomorphism is a mapping $h : \Sigma^* \rightarrow \Sigma^*$ with the property that for any $w_1, \dots, w_n \in \Sigma$ that $h(w_1w_2\dots w_n) = h(w_1)h(w_2)\dots h(w_n)$. We can define it for languages naturally as $h(L) = \{h(w) \mid w \in L\}$. Prove regular languages are closed under homomorphism.
3. Let L be any language. Define $L' = \{w \in \Sigma^* \mid wx \in L, x \in \Sigma^*\}$. Prove that if L is regular, then so is L' . This language is essentially the language of prefixes of L .
4. Prove that if L is regular, then so is $L \setminus \{a, bb, aba, bbab, bbaaa, ba\}$
5. Prove that $\{1^k \mid k \equiv 0 \pmod{n}\}$ cannot be done fewer than n states.
6. Prove every infinite regular language is the union of two disjoint other infinite regular languages.
7. Prove the following are not regular by use of the pumping lemma.
 - (a) Prove $\{a^mb^n \mid m, n \in \mathbb{N}, m \geq n\}$
 - (b) $\{w \in \Sigma^* \mid w \text{ is not a palindrome}\}$
 - (c) $\{u1v \in \Sigma^* \mid u, v \in \Sigma^* \text{ and } |u| = |v|\}$
 - (d) $\{0^n1^m \mid \gcd(n, m) = 1\}$
 - (e) $\{1^n01^n \mid n \in \mathbb{N}\}$
 - (f) $\{1^{n!} \mid n \in \mathbb{N}\}$
 - (g) $\{1^{2^n} \mid n \in \mathbb{N}\}$
 - (h) $\{0^{a_1}10^{a_2}1\dots10^{a_k} \mid i \neq j \implies a_i \neq a_j\}$
 - (i) $\{x + y = z \mid x, y, z \in (0 \cup 0)^*(0 \cup 1) \text{ and } \text{int}(x) + \text{int}(y) = \text{int}(z)\}$
 - (j) $\{1^n \mid n \text{ is composite}\}$
8. Let $\Sigma = \{0, 1, \#\}$ Prove
$$\{x_1\#x_2\#\dots\#x_k\#\#x \mid \text{each } x_i \text{ is of length } n \text{ and contains only } 0, 1 \text{ and } \exists i(x = x_i)\}$$
is not only regular, but requires 2^{2^n} states
9. Give an NFA, DFA, and regular expression for the following languages.

- (a) $\{w \in \Sigma^* \mid w \text{ ends with } abba\}$
 - (b) $\{w \in \Sigma^* \mid w \text{ has a single } b \text{ in the 3rd place from the right}\}$
10. (a) The method of ripping is not the earliest proof that $\mathcal{L}(NFA) \subseteq \mathcal{L}(REX)$. Let $D = (Q, \Sigma, q_0, \delta, F)$ be a DFA. We prove there is a regular expression R such that $L(R) = L(D)$. Let $R(i, j, k)$ be the set of strings which may go from q_i to q_j in D only using states $q_0, \dots, q_{k-1}$.
- i. Prove that $R(i, j, 1)$ has a regular expression to describe it.
 - ii. Suppose our induction hypothesis, that for all i, j and fixed k that $R(i, j, k)$ has a regular expression. Prove that

$$R(i, j, k) \cup R(i, k, k)R(k, k, k)^*R(k, j, k)$$

is a regular expression to describe $R(i, j, k+1)$

- iii. Prove that $L(D) = \bigcup_{q_j \in F} R(0, j, n+1)$ and conclude there is a regular expression to describe the language decided by D .
- (b) All we are doing is creating a regular expression to describe the strings which can bring a DFA from the start state to an accept state. The regular expression describes the valid paths. Richard Floyd saw this, and thought, hey I could use that to compute All-pairs Shortest Paths in directed graphs, and thus, gave us the Floyd-Warshall algorithm. The all-pairs-shortest-path problem is as follows. Your input is a directed edgeweighted graph $G = (V, E)$ with no negative cycles. You must output for each $v_i, v_j \in V$ a number $dist[i, j]$ which is the shortest path in G from v_i to v_j . Extend the NFA to regular expression algorithm to solve the all-pairs-shortest-path problem (Hint, instead of unions and concatenations, think about additions and minimums).

Let L be a language and its rotation $rot(L) = \{yx \mid xy \in L\}$. Prove that if L is regular then so is its rotation.

- 11. Prove $\{ww^Rx \mid w, x \in \Sigma^* \text{ and are not the empty string}\}$ is not regular without pumping. Throw closure at it.
- 12. Give a language which can be pumped but is not regular.
- 13. For each of the following, give the CFG, give the PDA
 - (a) $\emptyset$
 - (b) $\{a^n cb^n \mid n \in \mathbb{N}\}$
 - (c) aa^*b^* (or additionally, any other regular expression)
 - (d) $\{w \in \Sigma^* \mid \#a(w) = 2\#b(w)\}$
 - (e) $\{a^i b^j \mid i < j < 2i \in \mathbb{N}\}$
 - (f) $\{w \in \Sigma^* \mid |w| \text{ is odd}\}$
 - (g) $\overline{\{a^n b^n \mid n \in \mathbb{N}\}}$
 - (h) $\{xa^n yb^m zc^m y^R d^n x^R \mid n, m \in \mathbb{N} \text{ and } x, y, z \in \Sigma^*\}$

- (i) Strings of valid, balanced parenthesis.
- (j) valid arithmetical expressions on a calculator over $\Sigma = \{\times, +, -, (,), 0, \dots, 9\}$
- (k) Describe the infinitely many CFGs for Σ^*
 - (l) $\{w \in \Sigma^* \mid \#a(w) + \#b(w) = \#c(w) + \#d(w)\}$
 - (m) $\{ww^R \mid \#a(w) = 1\}$
 - (n) $\{a^ib^jc^k \mid i > j \text{ or } i + k < j\}$
- 14. Prove that every regular language is context-free. How many unique ways can you prove it?
- 15. Prove that the context-free languages are closed under union, Kleene star, concatenation, but NOT closed under intersection and complement. You may assume that $\{a^n b^n c^n \mid n \in \mathbb{N}\}$ is not context-free.
- 16. Give an algorithm on input a CFG G to determine if $L(G) = \emptyset$.
- 17. Challenge problem: Let L be a unary context-free language. Prove that L is the finite union of languages whose lengths are an arithmetic expression and a finite language. (I would be very interested if you found a solution. This is an generalization of your homework problem. It is known to be true, a corollary of Parikh's theorem. Can you prove it without Parikh's? By your homework problem, this will also prove every unary context-free language must be regular.)